

HW2 ML, Solution

Q1.

1.1.

The choice of the bandwidth "hn" is crucial in the Parzen method. A small bandwidth can lead to a high-variance estimate (sensitive to noise), while a large bandwidth can result in a high bias (oversmoothing and missing fine details). Therefore, there's a bias-variance tradeoff in selecting an appropriate bandwidth.

In KNN, The choice of the value for "kn" (k) also involves a bias-variance tradeoff. A small value of k (e.g., k=1) can lead to a low bias but high variance, as the decision is based on a single neighbor and is therefore sensitive to noise. A large value of k (e.g., k=n, where n is the total number of data points) can result in a smoother decision boundary, reducing variance but potentially introducing bias by oversimplifying the model.

1.2.

It is straightforward. In *K nearest neighbours (KNN)*, when we want to estimate probability density at a point $\mathbf{x}_i$, we will consider a small sphere centered on $\mathbf{x}_i$ and then allow the radius to grow until it contains K data points, and then $p(\mathbf{x}_i)$ will equal to $K/(NV_i)$, where N is total observations and V_i is the volume of the sphere centered on $\mathbf{x}_i$. We can assume that V_i is small enough that $p(\mathbf{x}_i)$ is roughly constant in it. In this way, We can write down the integral:

$$\int p(\mathbf{x}) d\mathbf{x} \approx \sum_{i=1}^N p(\mathbf{x}_i) \cdot V_i = \sum_{i=1}^N \frac{K}{NV_i} \cdot V_i = K \neq 1$$

We also see that if we use "*1NN*" ($K = 1$), the probability density will be well normalized. Note that if and only if the volume of all the spheres are small enough and N is large enough, the equation above will hold. Fortunately, these two conditions can be satisfied in *KNN*.

Q2.

$$p(x) = \begin{cases} 1/a & 0 \leq x \leq a \\ 0 & \text{otherwise,} \end{cases}$$

and our Parzen window is

$$\varphi(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

(a) The expected value of the Parzen window estimate is

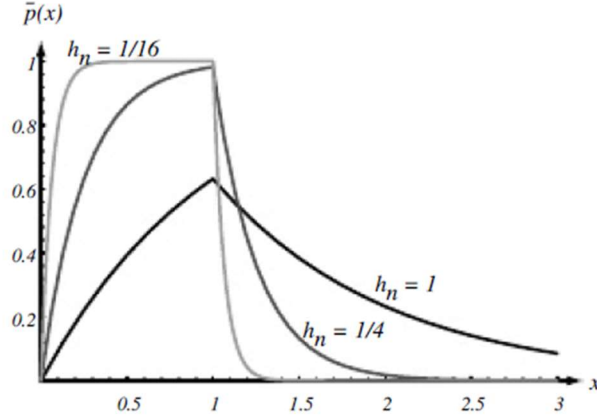
$$\begin{aligned} \bar{p}_n(x) &= \mathcal{E} \left[\frac{1}{nh_n} \sum_{i=1}^n \varphi \left(\frac{x - x_i}{h_n} \right) \right] = \frac{1}{h_n} \int \varphi \left(\frac{x - v}{h_n} \right) p(v) dv \\ &= \frac{1}{h_n} \int_{x \geq v} e^{-\frac{(x-v)}{h_n}} p(v) dv \\ &= \frac{\exp[\frac{x}{h_n}]}{h_n} \int_{\substack{x \geq v, \\ 0 < v < a}} \frac{1}{a} \exp[-v/h_n] dv \\ &= \begin{cases} 0 & \text{if } x < 0 \\ \frac{e^{-x/h_n}}{ah_n} \int_0^x e^{v/h_n} dv & \text{if } 0 \leq x \leq a \\ \frac{e^{-x/h_n}}{ah_n} \int_0^a e^{v/h_n} dv & \text{if } x \geq a \end{cases} \\ &= \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{a} (1 - e^{-x/h_n}) & \text{if } 0 \leq x \leq a \\ \frac{1}{a} (e^{a/h_n} - 1) e^{-x/h_n} & \text{if } x \geq a, \end{cases} \end{aligned}$$

where in the first step we used the fact that $x_1, \dots, x_n$ are independent samples drawn according to $p(v)$.

(b) For the case $a = 1$, we have

$$\bar{p}_n(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x/h_n} & 0 \leq x \leq 1 \\ (e^{1/h_n} - 1)e^{-x/h_n} & x > 1, \end{cases}$$

as shown in the figure.



(c) The bias is

$$\begin{aligned} p(x) - \bar{p}_n(x) &= \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{a} - \frac{1}{a}(1 - e^{-x/h_n}) & \text{if } 0 \leq x \leq a \\ 0 - \frac{1}{a}(e^{a/h_n} - 1)e^{-x/h_n} & \text{if } x \geq a \end{cases} \\ &= \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{a}e^{-x/h_n} & \text{if } 0 \leq x \leq a \\ -\frac{1}{a}(e^{a/h_n} - 1)e^{-x/h_n} & \text{if } x \geq a. \end{cases} \end{aligned}$$

Formally, a bias lower than 1% over 99% of the range $0 < x < a$, means that

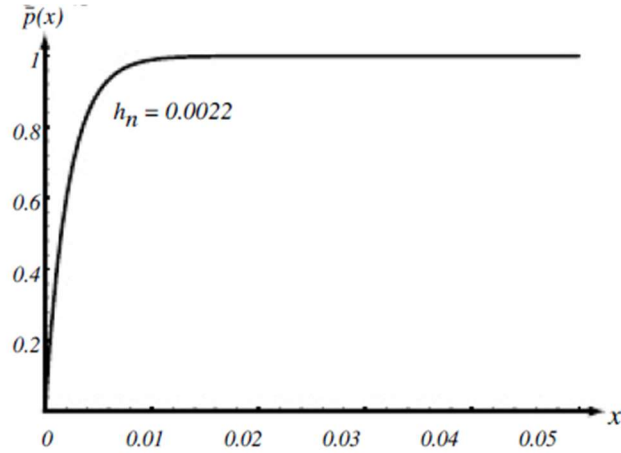
$$\frac{p(x) - \bar{p}(x)}{p(x)} \leq 0.01$$

over 99% of the range $0 < x < a$. This, in turn, implies

$$\frac{1/ae^{-x/h_n}}{1/a} \leq 0.01 \quad \text{over 99\% of } 0 < x < a \quad \text{or}$$

$$h_n \leq \frac{0.01a}{\ln(100)}.$$

For the case $a = 1$, we have that $h_n \leq 0.01/(\ln 100) = 0.0022$, as shown in the figure. Notice that the estimate is within 1% of $p(x) = 1/a = 1.0$ above $x \sim 0.01$, fulfilling the conditions of the problem.



Q3.

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

and our estimate is

$$p_n(x) = \frac{1}{nh_n} \sum_{i=1}^n \varphi\left(\frac{x-x_i}{h_n}\right).$$

- (a) The expected value of the probability at x , based on a window width parameter h_n , is

$$\begin{aligned} \bar{p}_n(x) &= \mathcal{E}[p_n(x)] = \frac{1}{nh_n} \sum_{i=1}^n \mathcal{E}\left[\varphi\left(\frac{x-x_i}{h_n}\right)\right] \\ &= \frac{1}{h_n} \int_{-\infty}^{\infty} \varphi\left(\frac{x-v}{h_n}\right) p(v) dv \\ &= \frac{1}{h_n} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-v}{h_n}\right)^2\right] \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2\right] dv \\ &= \frac{1}{2\pi h_n \sigma} \exp\left[-\frac{1}{2}\left(\frac{x^2}{h_n^2} + \frac{\mu^2}{\sigma^2}\right)\right] \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}v^2\left(\frac{1}{h_n^2} + \frac{1}{\sigma^2}\right) - 2v\left(\frac{x}{h_n^2} + \frac{\mu}{\sigma^2}\right)\right] dv \\ &= \frac{1}{2\pi h_n \sigma} \exp\left[-\frac{1}{2}\left(\frac{x^2}{h_n^2} + \frac{\mu^2}{\sigma^2}\right) + \frac{1}{2}\frac{\alpha^2}{\theta^2}\right] \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\frac{(v-\alpha)^2}{\theta^2}\right] dv, \end{aligned}$$

where we have defined $\left[\begin{pmatrix} \frac{x}{h_n^2} + \frac{\mu}{\sigma^2} \\ \frac{1}{h_n^2} + \frac{1}{\sigma^2} \end{pmatrix} \right]^{-1} = \begin{pmatrix} \alpha \\ \theta^2 \end{pmatrix}$

$$\theta^2 = \frac{1}{1/h_n^2 + 1/\sigma^2} = \frac{h_n^2 \sigma^2}{h_n^2 + \sigma^2}$$

and

$$\alpha = \theta^2 \left(\frac{x}{h_n^2} + \frac{\mu}{\sigma^2} \right).$$

We perform the integration and find

$$\begin{aligned}\bar{p}_n(x) &= \frac{\sqrt{2\pi}\theta}{2\pi h_n \sigma} \exp \left[-\frac{1}{2} \left(\frac{x^2}{h_n^2} + \frac{\mu^2}{\sigma^2} \right) + \frac{1}{2} \frac{\alpha^2}{\theta^2} \right] \\ &= \frac{1}{\sqrt{2\pi} h_n \sigma} \frac{h_n \sigma}{\sqrt{h_n^2 + \sigma^2}} \exp \left[-\frac{1}{2} \left(\frac{x^2}{h_n^2} + \frac{\mu^2}{\sigma^2} - \frac{\alpha^2}{\theta^2} \right) \right].\end{aligned}$$

The argument of the exponentiation can be expressed as follows

$$\begin{aligned}\frac{x^2}{h_n^2} + \frac{\mu^2}{\sigma^2} - \frac{\alpha^2}{\theta^2} &= \frac{x^2}{h_n^2} + \frac{\mu^2}{\sigma^2} - \frac{\theta^4}{\theta^2} \left(\frac{x}{h_n^2} + \frac{\mu}{\sigma^2} \right)^2 \\ &= \frac{x^2 h_n^2}{(h_n^2 + \sigma^2) h_n^2} + \frac{\mu^2 \sigma^2}{(h_n^2 + \sigma^2) \sigma^2} - \frac{2x\mu}{h_n^2 + \sigma^2} \\ &= \frac{(x - \mu)^2}{h_n^2 + \sigma^2}.\end{aligned}$$

We substitute this back to find

$$\bar{p}_n(x) = \frac{1}{\sqrt{2\pi} \sqrt{h_n^2 + \sigma^2}} \exp \left[-\frac{1}{2} \frac{(x - \mu)^2}{h_n^2 + \sigma^2} \right],$$

which is the form of a Gaussian, denoted

$$\bar{p}_n(x) = N(\mu, h_n^2 + \sigma^2).$$

(b) We calculate the variance as follows:

$$\begin{aligned}\text{Var}[p_n(x)] &= \text{Var} \left[\frac{1}{nh_n} \sum_{i=1}^n \varphi \left(\frac{x - x_i}{h_n} \right) \right] \\ &= \frac{1}{n^2 h_n^2} \sum_{i=1}^n \text{Var} \left[\varphi \left(\frac{x - x_i}{h_n} \right) \right] \\ &= \frac{1}{nh_n^2} \text{Var} \left[\varphi \left(\frac{x - v}{h_n} \right) \right] \\ &= \frac{1}{nh_n^2} \left\{ \mathcal{E} \left[\varphi^2 \left(\frac{x - v}{h_n} \right) \right] - \left(\mathcal{E} \left[\varphi \left(\frac{x - v}{h_n} \right) \right] \right)^2 \right\},\end{aligned}$$

where in the first step we used the fact that $\mathbf{x}_1, \dots, \mathbf{x}_n$ are independent samples drawn according to $p(\mathbf{x})$. We thus now need to calculate the expected value of the square of the kernel function

$$\begin{aligned}\mathcal{E} \left[\varphi^2 \left(\frac{x - v}{h_n} \right) \right] &= \int \varphi^2 \left(\frac{x - v}{h_n} \right) p(v) dv \\ &= \int_{-\infty}^{\infty} \frac{1}{2\pi} \exp \left[-\left(\frac{x - v}{h_n} \right)^2 \right] \exp \left[-\frac{1}{2} \left(\frac{v - \mu}{\sigma} \right)^2 \right] \frac{1}{\sqrt{2\pi}\sigma} dv \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - v}{h_n/\sqrt{2}} \right)^2 \right] \exp \left[-\frac{1}{2} \left(\frac{v - \mu}{\sigma} \right)^2 \right] \frac{1}{\sqrt{2\pi}\sigma} dv.\end{aligned}$$

From part (a) by a similar argument with $h_n/\sqrt{2}$ replacing h_n , we have

$$\begin{aligned} \frac{1}{h_n/\sqrt{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-v}{h_n/\sqrt{2}} \right)^2 \right] \exp \left[-\frac{1}{2} \left(\frac{v-\mu}{\sigma} \right)^2 \right] \frac{1}{\sqrt{2\pi}\sigma} dv \\ = \frac{1}{\sqrt{2\pi}} \frac{1}{h_n^2/2 + \sigma^2} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2/2 + \sigma^2} \right]. \end{aligned}$$

We make the substitution and find

$$\mathcal{E} \left[\varphi^2 \left(\frac{x-v}{h_n} \right) \right] = \frac{h_n/\sqrt{2}}{2\pi\sqrt{h_n^2/2 + \sigma^2}} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2/2 + \sigma^2} \right],$$

and thus conclude

$$\frac{1}{nh_n^2} \mathcal{E} \left[\varphi^2 \left(\frac{x-v}{h_n} \right) \right] = \frac{1}{nh_n} \frac{1}{2\sqrt{\pi}} \frac{1}{\sqrt{2\pi}\sqrt{h_n^2/2 + \sigma^2}} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2/2 + \sigma^2} \right].$$

For small h_n , $\sqrt{h_n^2/2 + \sigma^2} \simeq \sigma$, and thus the above equation can be approximated as

$$\begin{aligned} \frac{1}{nh_n^2} \mathcal{E} \left[\varphi^2 \left(\frac{x-v}{h_n} \right) \right] &\simeq \frac{1}{nh_n 2\sqrt{\pi}} \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right] \\ &= \frac{1}{2nh_n\sqrt{\pi}} p(x). \end{aligned} \quad (*)$$

Similarly, we have

$$\begin{aligned} \frac{1}{nh_n^2} \left\{ \mathcal{E} \left[\varphi \left(\frac{x-v}{h_n} \right) \right] \right\}^2 &= \frac{1}{nh_n^2} h_n^2 \frac{1}{\sqrt{2\pi}\sqrt{h_n^2 + \sigma^2}} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2 + \sigma^2} \right] \\ &= \frac{h_n}{nh_n \sqrt{2\pi}\sqrt{h_n^2 + \sigma^2}} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2 + \sigma^2} \right] \\ &\simeq \frac{h_n}{nh_n \sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right] \simeq 0, (**) \end{aligned}$$

valid for small h_n . From (*) and (**) we have, (still for small h_n)

$$\text{Var}[P_n(x)] \simeq \frac{p(x)}{2nh_n\sqrt{\pi}}.$$

(c) We write the bias as

$$\begin{aligned} p(x) - \bar{p}^n(x) &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right] - \frac{1}{\sqrt{2\pi}\sqrt{h_n^2 + \sigma^2}} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2 + \sigma^2} \right] \\ &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{h_n^2 + \sigma^2} \right] \left\{ 1 - \frac{\sigma}{\sqrt{h_n^2 + \sigma^2}} \exp \left[\frac{1}{2} \frac{(x-\mu)^2}{h_n^2 + \sigma^2} - \frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right] \right\} \\ &= p(x) \left\{ 1 - \frac{1}{\sqrt{1 + (h_n/\sigma)^2}} \exp \left[-\frac{(x-\mu)^2}{2} \left\{ \frac{1}{h_n^2 + \sigma^2} - \frac{1}{\sigma^2} \right\} \right] \right\} \\ &= p(x) \left\{ 1 - \frac{1}{\sqrt{1 + (h_n/\sigma)^2}} \exp \left[\frac{1}{2} \frac{h_n^2 (x-\mu)^2}{h_n^2 + \sigma^2} \right] \right\}. \end{aligned}$$

For small h_n we expand to second order:

$$\frac{1}{\sqrt{1 + \left(\frac{h_n}{\sigma}\right)^2}} \simeq 1 - \frac{1}{2} (h_n/\sigma)^2$$

and

$$\exp \left[\frac{h_n^2 (x - \mu)^2}{2\sigma^2 h_n^2 + \sigma^2} \right] \simeq 1 + \frac{h_n^2 (x - \mu)^2}{2\sigma^2 h_n^2 + \sigma^2}.$$

We ignore terms of order higher than h_n^2 and find

$$\begin{aligned} p(x) - \bar{p}_n(x) &\simeq p(x) \left\{ 1 - \left(1 - \frac{1}{2} \left(\frac{h_n}{\sigma} \right)^2 \right) \left(1 + \frac{h_n^2 (x - \mu)^2}{2\sigma^2 h_n^2 + \sigma^2} \right) \right\} \\ &\simeq p(x) \left\{ 1 - 1 + \frac{1}{2} \frac{h_n^2}{\sigma^2} - \frac{h_n^2 (x - \mu)^2}{2\sigma^2 h_n^2 + \sigma^2} \right\} \\ &\simeq \frac{1}{2} \left(\frac{h_n}{\sigma} \right)^2 \left[1 - \frac{(x - \mu)^2}{h_n^2 + \sigma^2} \right] p(x) \\ &\simeq \frac{1}{2} \left(\frac{h_n}{\sigma} \right)^2 \left[1 - \left(\frac{x - \mu}{\sigma} \right)^2 \right] p(x). \end{aligned}$$

Q4.

1)

By applying Jensen inequality we have:

$$p(\theta|x) \propto p(x|\theta)p(\theta) \propto \left(\sum_z Q(z) \frac{p(x, z|\theta)}{Q(z)} \right) p(\theta) \implies$$

$$L = \ln \left(\sum_z Q(z) \frac{p(x, z|\theta)}{Q(z)} \right) + \ln(p(\theta)) \implies L \geq \sum_z Q(z) \{ \ln(p(x, z|\theta)) - \ln(Q(z)) \} + \ln(p(\theta))$$

for E-step we have to maximize above function by $Q(z)$

$$\frac{\partial L}{\partial Q(j)} = \ln(p(z, x|\theta)) - \ln(Q(j)) - 1 = 0 \implies Q(j) = Cp(z = j, x|\theta)$$

we have to normalize respect to C

$$\begin{aligned} \sum_z Cp(z = z, x|\theta) = 1 &\implies C = \frac{1}{\sum_z p(z = z, x|\theta)} \implies Q(z) = \frac{p(z, x|\theta)}{\sum_{k=1}^Z p(z = k, x|\theta)} \\ Q(z) &= \frac{p(z, x|\theta)}{p(x|\theta)} = p(z|x, \theta) \checkmark \end{aligned}$$

For M-step we have to maximize L respect to θ so :

$$\begin{aligned}
& \text{maximize} \sum_z Q(z) \{ \ln(p(x, z|\theta)) - \ln(Q(z)) \} + \ln(p(\theta)) = \\
& \text{maximize} \sum_z Q(z) \ln(p(x, z|\theta)) - \sum_z Q(z) \ln(Q(z)) + \ln(p(\theta)) = \\
& \text{maximize} \sum_z Q(z) \ln(p(x, z|\theta)) + \text{const} + \ln(p(\theta)) \propto \text{maximize} \sum_z Q(z) \ln(p(x, z|\theta)) + \ln(p(\theta))
\end{aligned}$$

so as you see the only difference is $Q(\theta^{old}, \theta) = Q(\theta^{old}) + \ln(p(\theta))$ then we have:

$$\begin{aligned}
& \text{maximize}_{\theta} \sum_z Q_{\theta^{old}}(z) \ln(p(x, z|\theta)) + \ln(p(\theta)) \\
& Q(z) = p(z|x, \theta) \\
& \text{maximize}_{\theta} \sum_z p(z|x, \theta^{old}) \ln(p(x, z|\theta)) + \ln(p(\theta)) \checkmark
\end{aligned}$$

2)

E-step

As it's said in problem we don't have n_a & n_b so we will only calculate E-step for this two labels.

as we know for two label B & D the number are given so prior probability for these two label are 1 or zero for each data as the label of these two are given:

$$\begin{aligned}
p(B|\theta, x) &= 1 \text{ or } 0 \\
p(D|\theta, x) &= 1 \text{ or } 0 \\
&\text{depends on data labels}
\end{aligned}$$

but for two other labels we got to calculate prior as below:

$$\begin{aligned}
p(A|\theta, x) &= \frac{p(A, x|\theta)}{\sum_{k \in \{A, C\}} p(z = k, x|\theta)} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{2\theta}{3}} = \frac{1}{1 + 2\theta} \\
p(C|\theta, x) &= \frac{p(C, x|\theta)}{\sum_{k \in \{A, C\}} p(z = k, x|\theta)} = \frac{\frac{2\theta}{3}}{\frac{1}{3} + \frac{2\theta}{3}} = \frac{2\theta}{1 + 2\theta} \\
&\forall x \in \text{incomplete data}
\end{aligned}$$

M-step

first consider $\beta = n_a + n_c = N - n_d - n_b$ then we have:

$$\begin{aligned}
 & \text{argmax}_{\theta} L(\theta) \\
 L &= \sum_N \sum_Z p(z|\theta^{old}, x_n) \ln(p(x, z|\theta)) + \ln(p(\theta)) = \\
 & n_b \ln\left(\frac{1}{3}(1-\theta)\right) + n_d \ln\left(\frac{1}{3}(1-\theta)\right) + \sum_N \sum_{z \in \{A, C\}} p(z|\theta^{old}, x_n) \ln(p(x, z|\theta)) + \ln(p(\theta)) = \\
 & n_b \ln\left(\frac{1}{3}(1-\theta)\right) + n_d \ln\left(\frac{1}{3}(1-\theta)\right) + \sum_N \sum_{z \in \{A, C\}} p(z|\theta^{old}, x_n) \ln(p(x, z|\theta)) + \ln(p(\theta)) = \\
 & n_b \ln\left(\frac{1}{3}(1-\theta)\right) + n_d \ln\left(\frac{1}{3}(1-\theta)\right) + \beta \left(\frac{1}{1+2\theta^{old}} \ln\left(\frac{1}{3}\right) + \frac{2\theta^{old}}{1+2\theta^{old}} \ln\left(\frac{2\theta}{3}\right) \right) + \\
 & \ln(\Gamma(v_1 + v_2)) - \ln(\Gamma(v_1)) - \ln(\Gamma(v_2)) + (v_1 - 1) \ln(\theta) + (v_2 - 1) \ln((1 - \theta)) \implies \\
 & \frac{\partial L}{\partial \theta} = \frac{-(n_b + n_d)}{1 - \theta} + \beta \frac{2\theta^{old}}{1 + 2\theta^{old}} \frac{1}{\theta} + \frac{v_1 - 1}{\theta} + \frac{-(v_2 - 1)}{1 - \theta} = 0 \\
 & \theta^{new} = \frac{v_1 - 1 + \beta \frac{2\theta^{old}}{1 + 2\theta^{old}}}{v_1 - 1 + \beta \frac{2\theta^{old}}{1 + 2\theta^{old}} + v_2 - 1 + n_b + n_d} \checkmark \\
 & \text{where } \beta = n_a + n_c = N - n_d - n_b
 \end{aligned}$$

Q5.

1)

As it's binomial distribution it's easy to obtain:

$$\begin{aligned}
 I) \quad p(w_t, m_t | k) &= \binom{m_t}{w_t} p_k^{w_t} (1 - p_k)^{m_t - w_t} \\
 II) \quad p(w_t, m_t | ct) &= \binom{m_t}{w_t} p_{ct}^{w_t} (1 - p_{ct})^{m_t - w_t}
 \end{aligned}$$

2)

As we know from previous question for E-step we have :

$$Q(k) = p(k|x, \theta) \propto p(x|ct = k, \theta)p(ct = k) \propto \binom{m_t}{x} p_k^x (1 - p_k)^{m_t - x} \pi_k$$

$$\sum_K Q(k) = 1 \implies \sum_K \binom{m_t}{x} p_k^x (1 - p_k)^{m_t - x} \pi_k = 1$$

As we know the equation above is propto so we need to normalize it that it adds up to one:

$$Q_t[k]^i = \frac{\binom{m_t}{x} p_k^x (1 - p_k)^{m_t - x} \pi_k}{\sum_K \binom{m_t}{x} p_j^x (1 - p_j)^{m_t - x} \pi_j}$$

3)

For writing M-step we have to maximize equation below:

$$L = \sum_N \sum_K Q_t[k]^i \log(p(w_t|p_k)) = \sum_N \sum_K Q_t[k]^i \log\left(\binom{m_t}{w_t} p_k^{w_t} (1 - p_k)^{m_t - w_t}\right) =$$

$$\sum_N \sum_K Q_t[k]^i \left\{ \log\left(\binom{m_t}{w_t}\right) + w_t \log(p_k) + (m_t - w_t) \log(1 - p_k) \right\}$$

$$\frac{\partial L}{\partial p_j} = \sum_N Q_t[j]^i \left\{ \frac{w_t}{p_j} - \frac{m_t - w_t}{1 - p_j} \right\} = \sum_N Q_t[j]^i \left\{ \frac{w_t - m_t p_j}{p_j - p_j^2} \right\} = 0 \implies$$

$$\sum_N Q_t[j]^i \{w_t - m_t p_j\} = 0 \implies p_j = \frac{\sum_{i=1}^N Q_t[j]^i w_t}{\sum_{i=1}^N Q_t[j]^i m_t} \checkmark$$